



Solution Set

Question 1

Consider a European equity option under geometric Brownian motion. Define the state variables (spot, strike, rate, dividend yield, volatility, maturity, and Greeks) and derive the Black-Scholes PDE and closed-form call value. State the measure, filtration, and tradability assumptions.

$$\frac{\partial V}{\partial t} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} + rS \frac{\partial V}{\partial S} - rV = 0$$

Answer key

Define the tradable state variables first: spot, strike, rate, dividend yield, volatility, maturity, and Greeks. The payoff is then written at the relevant horizon and valued under the pricing measure.

The derivation uses the economic relation for Black-Scholes PDE and closed-form call value. If a hedge or replication argument is used, the stochastic exposure is neutralized locally and the remaining cash account earns the risk-free or collateral rate.

The final expression must agree with the payoff at maturity and with the obvious limiting cases.

$$\begin{aligned} \frac{\partial V}{\partial t} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} + rS \frac{\partial V}{\partial S} - rV &= 0 \\ d\Pi &= dV - \Delta dS, \quad d\Pi = r\Pi dt \end{aligned}$$

Question 2

For a European equity option under geometric Brownian motion, explain which probability measure is natural for pricing. Write the density-process logic needed if the starting model is under the physical measure.

Answer key

The pricing measure is chosen so discounted tradable prices are martingales under the chosen numeraire.

The Radon-Nikodym density reweights probabilities; it does not change the contractual payoff. In Brownian models, Girsanov changes the drift and leaves the quadratic variation unchanged.

A useful check is that the density has expectation one and that the resulting discounted price process has zero drift.

$$\begin{aligned} \frac{\partial V}{\partial t} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} + rS \frac{\partial V}{\partial S} - rV &= 0 \\ Z_t = \frac{dQ}{dP} \Big|_{\mathcal{F}_t}, \quad E^Q[X | \mathcal{F}_t] &= \frac{1}{Z_t} E^P[Z_T X | \mathcal{F}_t] \end{aligned}$$

Question 3

Identify the discounted tradable or density-adjusted process in a European equity option under geometric Brownian motion. Prove the martingale condition and name one modelling choice that would break it.

Answer key

The martingale candidate is the discounted tradable price, or the density-adjusted process after a measure change.

Condition on the earlier information set and verify that the conditional expectation equals the earlier value. Any remaining predictable drift indicates a missing funding, dividend, numeraire, or model adjustment.

For this topic, the practical issue is using physical drift in an arbitrage-free price.

$$\begin{aligned} \frac{\partial V}{\partial t} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} + rS \frac{\partial V}{\partial S} - rV &= 0 \\ E[M_t | \mathcal{F}_s] &= M_s, \quad 0 \leq s < t \end{aligned}$$

Question 4

Using spot 100, strike 100, rate 4 percent, dividend 1 percent, volatility 22 percent, maturity one year, construct a two-step numerical calculation. Show intermediate state values before discounting or aggregating.

$$\frac{\partial V}{\partial t} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} + rS \frac{\partial V}{\partial S} - rV = 0$$

Answer key

Use the stated market inputs: spot 100, strike 100, rate 4 percent, dividend 1 percent, volatility 22 percent, maturity one year. Compute future state values first, then take the correct conditional average, then discount.

Keep a calculation table with states, probabilities, payoffs, discount factors, and final present values. This makes unit mistakes visible.

The answer should include a sign and magnitude check, not only a final number.

$$\frac{\partial V}{\partial t} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} + rS \frac{\partial V}{\partial S} - rV = 0$$

$$V_t = e^{-r\Delta t}(qV_u + (1-q)V_d)$$

Question 5

For a European equity option under geometric Brownian motion, compute and interpret delta, gamma, theta, vega, and PDE residual. Explain the hedge sign and the unit of each sensitivity.

Answer key

The relevant exposure is delta, gamma, theta, vega, and PDE residual. Compute the first-order term, then add convexity or cross terms only when the state variables require them.

A hedge has the opposite sign of the exposure. Delta is per unit of underlying, DV01 is per basis point, and vega is per volatility point or percent depending on convention.

After hedging, report the residual risk that remains.

$$\frac{\partial V}{\partial t} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} + rS \frac{\partial V}{\partial S} - rV = 0$$

$$dV \approx \Delta dS + \frac{1}{2}\Gamma(dS)^2 + \nu d\sigma$$

Question 6

Suppose the desk observes implied volatility from a quoted option premium. Write the calibration objective, the weights you would use, and the parameter-identifiability checks.

Answer key

The calibration instruments are implied volatility from a quoted option premium. Write a weighted least-squares objective using market quotes and model quotes.

Weights should reflect liquidity, bid-ask width, and quote reliability. Parameter constraints should prevent impossible prices or unstable extrapolation.

After calibration, inspect residuals and check whether each parameter has a distinct effect on the quotes.

$$\frac{\partial V}{\partial t} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} + rS \frac{\partial V}{\partial S} - rV = 0$$

$$\hat{\theta} = \arg \min_{\theta} \sum_i w_i (M_i(\theta) - Q_i^{\text{mkt}})^2$$

Question 7

State the terminal and boundary conditions for a European equity option under geometric Brownian motion. Explain how terminal payoff and low/high spot asymptotics prevents economically impossible prices.

$$\frac{\partial V}{\partial t} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} + rS \frac{\partial V}{\partial S} - rV = 0$$

Answer key

The terminal condition comes from the payoff. Boundary conditions come from economic limits such as zero spot, very large spot, barrier touch, exercise, or discount-factor positivity.

For this product, the key boundary idea is terminal payoff and low/high spot asymptotics.

A boundary condition is wrong if it permits arbitrage, negative values for non-negative payoffs, or a price that violates obvious monotonicity.

$$\frac{\partial V}{\partial t} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} + rS \frac{\partial V}{\partial S} - rV = 0$$

$$V(T, S) = g(S), \quad V(t, S_{\text{boundary}}) = \text{economic boundary value}$$

Question 8

Design a Monte Carlo or lattice estimator for a European equity option under geometric Brownian motion. Specify risk-neutral GBM paths benchmarked to the closed form, the estimator, the discounting rule, and the standard error.

Answer key

Simulate risk-neutral GBM paths benchmarked to the closed form. For each path, compute the payoff using only information available along that path.

The estimator is the discounted sample average. Report standard error and rerun with larger path counts or finer time steps to separate sampling error from discretization error.

For path-dependent products, store the running state variable explicitly.

$$\frac{\partial V}{\partial t} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} + rS \frac{\partial V}{\partial S} - rV = 0$$

$$\widehat{V}_0 = e^{-rT} \frac{1}{N} \sum_{i=1}^N X^{(i)}, \quad \widehat{\text{SE}} = e^{-rT} \frac{s_X}{\sqrt{N}}$$

Question 9

A junior quant implements a European equity option under geometric Brownian motion and gets a stable but wrong answer. Explain how using physical drift in an arbitrage-free price creates the error and design a diagnostic test.

Answer key

The main implementation risk is using physical drift in an arbitrage-free price. This can produce stable but wrong results because the error is systematic.

A good diagnostic compares the implementation with a limiting case or benchmark where the direction of the answer is known.

Change one input at a time and verify monotonicity, units, and convergence.

$$\frac{\partial V}{\partial t} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} + rS \frac{\partial V}{\partial S} - rV = 0$$

$$\text{error}(N, \Delta t) = \widehat{V}_{N, \Delta t} - V_{\text{benchmark}}$$

Question 10

Write an NumPy algorithm for a European equity option under geometric Brownian motion. Include inputs, outputs, diagnostics, and one benchmark case.

$$\frac{\partial V}{\partial t} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} + rS \frac{\partial V}{\partial S} - rV = 0$$

Answer key

Create clear arrays for inputs, states, probabilities, discount factors, and payoffs. Check dimensions before combining them.

Return both the main value and diagnostics such as convergence, residuals, standard error, or benchmark difference.

The routine is acceptable only if the result can be reproduced from the displayed assumptions and data.

$$\frac{\partial V}{\partial t} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} + rS \frac{\partial V}{\partial S} - rV = 0$$

$$\text{tree cost} = O(n^2), \quad \text{simulation cost} = O(NM)$$

Question 11

Starting from the payoff definition of a European equity option under geometric Brownian motion, derive a replication, dynamic-programming, or conditional-expectation representation without skipping the economic argument.

Answer key

Define the tradable state variables first: spot, strike, rate, dividend yield, volatility, maturity, and Greeks. The payoff is then written at the relevant horizon and valued under the pricing measure.

The derivation uses the economic relation for Black-Scholes PDE and closed-form call value. If a hedge or replication argument is used, the stochastic exposure is neutralized locally and the remaining cash account earns the risk-free or collateral rate.

The final expression must agree with the payoff at maturity and with the obvious limiting cases.

$$\frac{\partial V}{\partial t} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} + rS \frac{\partial V}{\partial S} - rV = 0$$

$$d\Pi = dV - \Delta dS, \quad d\Pi = r\Pi dt$$

Question 12

Create a stressed-market version of spot 100, strike 100, rate 4 percent, dividend 1 percent, volatility 22 percent, maturity one year. Recompute the value or risk metric and explain which term dominates the change.

Answer key

Use the stated market inputs: spot 100, strike 100, rate 4 percent, dividend 1 percent, volatility 22 percent, maturity one year. Compute future state values first, then take the correct conditional average, then discount.

Keep a calculation table with states, probabilities, payoffs, discount factors, and final present values. This makes unit mistakes visible.

The answer should include a sign and magnitude check, not only a final number.

$$\frac{\partial V}{\partial t} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} + rS \frac{\partial V}{\partial S} - rV = 0$$

$$V_t = e^{-r\Delta t}(qV_u + (1-q)V_d)$$

Question 13

Use a second-order Taylor approximation for a European equity option under geometric Brownian motion. Separate linear exposure, convexity exposure, cross terms if relevant, and residual model risk.

$$\frac{\partial V}{\partial t} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} + rS \frac{\partial V}{\partial S} - rV = 0$$

Answer key

The relevant exposure is delta, gamma, theta, vega, and PDE residual. Compute the first-order term, then add convexity or cross terms only when the state variables require them.

A hedge has the opposite sign of the exposure. Delta is per unit of underlying, DV01 is per basis point, and vega is per volatility point or percent depending on convention.

After hedging, report the residual risk that remains.

$$\begin{aligned} \frac{\partial V}{\partial t} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} + rS \frac{\partial V}{\partial S} - rV &= 0 \\ dV &\approx \Delta dS + \frac{1}{2}\Gamma (dS)^2 + \nu d\sigma \end{aligned}$$

Question 14

Show how a change of numeraire or change of measure would simplify one expectation in a European equity option under geometric Brownian motion. Identify what becomes a martingale after the change.

Answer key

The pricing measure is chosen so discounted tradable prices are martingales under the chosen numeraire.

The Radon-Nikodym density reweights probabilities; it does not change the contractual payoff. In Brownian models, Girsanov changes the drift and leaves the quadratic variation unchanged.

A useful check is that the density has expectation one and that the resulting discounted price process has zero drift.

$$\begin{aligned} \frac{\partial V}{\partial t} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} + rS \frac{\partial V}{\partial S} - rV &= 0 \\ Z_t = \frac{dQ}{dP} \Big|_{\mathcal{F}_t}, \quad E^Q[X | \mathcal{F}_t] &= \frac{1}{Z_t} E^P[Z_T X | \mathcal{F}_t] \end{aligned}$$

Question 15

Give a proof-style answer showing when the wealth process for a European equity option under geometric Brownian motion is self-financing. Explain why predictable trading strategies matter.

Answer key

The martingale candidate is the discounted tradable price, or the density-adjusted process after a measure change.

Condition on the earlier information set and verify that the conditional expectation equals the earlier value. Any remaining predictable drift indicates a missing funding, dividend, numeraire, or model adjustment.

For this topic, the practical issue is using physical drift in an arbitrage-free price.

$$\begin{aligned} \frac{\partial V}{\partial t} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} + rS \frac{\partial V}{\partial S} - rV &= 0 \\ E[M_t | \mathcal{F}_s] &= M_s, \quad 0 \leq s < t \end{aligned}$$

Question 16

Market quotes for implied volatility from a quoted option premium are noisy. Propose a robust calibration workflow with initial guesses, constraints, residual plots, and out-of-sample checks.

$$\frac{\partial V}{\partial t} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} + rS \frac{\partial V}{\partial S} - rV = 0$$

Answer key

The calibration instruments are implied volatility from a quoted option premium. Write a weighted least-squares objective using market quotes and model quotes.

Weights should reflect liquidity, bid-ask width, and quote reliability. Parameter constraints should prevent impossible prices or unstable extrapolation.

After calibration, inspect residuals and check whether each parameter has a distinct effect on the quotes.

$$\frac{\partial V}{\partial t} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} + rS \frac{\partial V}{\partial S} - rV = 0$$

$$\hat{\theta} = \arg \min_{\theta} \sum_i w_i (M_i(\theta) - Q_i^{\text{mkt}})^2$$

Question 17

For a European equity option under geometric Brownian motion, compare the mathematical boundary condition with the trader's intuition. Give one limiting case where the answer should be obvious.

Answer key

The terminal condition comes from the payoff. Boundary conditions come from economic limits such as zero spot, very large spot, barrier touch, exercise, or discount-factor positivity.

For this product, the key boundary idea is terminal payoff and low/high spot asymptotics.

A boundary condition is wrong if it permits arbitrage, negative values for non-negative payoffs, or a price that violates obvious monotonicity.

$$\frac{\partial V}{\partial t} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} + rS \frac{\partial V}{\partial S} - rV = 0$$

$$V(T, S) = g(S), \quad V(t, S_{\text{boundary}}) = \text{economic boundary value}$$

Question 18

Design a convergence study for risk-neutral GBM paths benchmarked to the closed form. Specify path counts, time-step refinement, confidence intervals, and the exact quantity plotted.

Answer key

Simulate risk-neutral GBM paths benchmarked to the closed form. For each path, compute the payoff using only information available along that path.

The estimator is the discounted sample average. Report standard error and rerun with larger path counts or finer time steps to separate sampling error from discretization error.

For path-dependent products, store the running state variable explicitly.

$$\frac{\partial V}{\partial t} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} + rS \frac{\partial V}{\partial S} - rV = 0$$

$$\widehat{V}_0 = e^{-rT} \frac{1}{N} \sum_{i=1}^N X^{(i)}, \quad \widehat{\text{SE}} = e^{-rT} \frac{s_X}{\sqrt{N}}$$

Question 19

Find one way that using physical drift in an arbitrage-free price could pass a superficial unit test. Write a stronger test that would fail for the biased implementation.

$$\frac{\partial V}{\partial t} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} + rS \frac{\partial V}{\partial S} - rV = 0$$

Answer key

The main implementation risk is using physical drift in an arbitrage-free price. This can produce stable but wrong results because the error is systematic.

A good diagnostic compares the implementation with a limiting case or benchmark where the direction of the answer is known.

Change one input at a time and verify monotonicity, units, and convergence.

$$\frac{\partial V}{\partial t} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} + rS \frac{\partial V}{\partial S} - rV = 0$$

$$\text{error}(N, \Delta t) = \hat{V}_{N, \Delta t} - V_{\text{benchmark}}$$

Question 20

Explain a European equity option under geometric Brownian motion and code the core routine. Give the derivation outline, pseudocode, and three numerical sanity checks.

Answer key

Create clear arrays for inputs, states, probabilities, discount factors, and payoffs. Check dimensions before combining them.

Return both the main value and diagnostics such as convergence, residuals, standard error, or benchmark difference.

The routine is acceptable only if the result can be reproduced from the displayed assumptions and data.

$$\frac{\partial V}{\partial t} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} + rS \frac{\partial V}{\partial S} - rV = 0$$

$$\text{tree cost} = O(n^2), \quad \text{simulation cost} = O(NM)$$